Unsupervised Learning

Complete Study Guide — Chapter 2

Clustering | Association Rules | K-Means | Apriori | ECLAT

Subject	Machine Learning — Unsupervised Methods
Level	Beginner to Intermediate
Sections	9 (Notes · Algorithms · Math · Visuals · Workflow · Applications · Quiz · Answers · Glossary)
Format	Conceptual + Mathematical + Visual

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Section 1

■ Conceptual Notes

Definitions · Analogies · Core Concepts

What is Unsupervised Learning?

Unsupervised learning is a type of machine learning where an algorithm is trained on **unlabeled data** — data without any pre-defined categories or correct answers. The algorithm must independently discover hidden patterns, structures, or groupings within the data. Unlike supervised learning, there is no 'teacher' guiding the model.

■ Intuition

■ ***Analogy:*** *Imagine you give a child a pile of 500 LEGO bricks of different colours, shapes, and sizes — with no instructions. The child naturally starts grouping them: red bricks here, blue there, flat ones in one pile, tall ones in another. The child discovered the structure on their own — that is unsupervised learning!*

Supervised vs Unsupervised Learning

Feature	Supervised Learning	Unsupervised Learning
Data	Labeled (has answers)	Unlabeled (no answers)
Goal	Predict output for new inputs	Find hidden patterns/groups
Feedback	Model learns from correct labels	No feedback, self-organized
Examples	Spam detection, price prediction	Customer segmentation, topic modeling
Algorithms	Decision Tree, SVM, Neural Net	K-Means, DBSCAN, Apriori
Evaluation	Accuracy, F1-Score, RMSE	Silhouette score, Inertia

Clustering

Clustering is the process of partitioning data points into groups (**clusters**) such that points within the same cluster are more similar to each other than to points in other clusters. Similarity is typically measured using **distance metrics**.

- Groups similar data points together automatically.
- Number of clusters may or may not be specified in advance.
- Used for customer segmentation, image compression, document grouping.
- Common algorithms: K-Means, K-Medoids, DBSCAN, Hierarchical Clustering.

Association Rules

Association rule learning discovers interesting **if–then relationships** between variables in large datasets. The most classic example is **Market Basket Analysis**: finding which products are frequently purchased together.

■ Intuition

■ ***Analogy:*** *A supermarket notices that customers who buy bread and butter very often also buy milk. This pattern — 'If bread AND butter, then milk' — is an association rule. The store uses this to place milk near bread!*

Section 2

■ Core Algorithms

K-Means · Elbow Method · K-Medoids · Apriori · ECLAT

K-Means Clustering

K-Means is the most widely used clustering algorithm. It partitions n data points into K clusters, where each point belongs to the cluster with the nearest mean (centroid). The algorithm iterates until centroids stabilize.

Step-by-Step Process:

#	Phase	Description
Step 1	Choose K	Decide the number of clusters K you want to create.
Step 2	Initialize	Randomly place K centroids in the data space.
Step 3	Assign	Assign each data point to the nearest centroid using Euclidean distance.
Step 4	Update	Recalculate each centroid as the mean of all points assigned to it.
Step 5	Repeat	Repeat Steps 3–4 until centroids do not change (convergence).
Step 6	Output	Final clusters are the result. Each centroid represents its cluster.

The Elbow Method

A key challenge in K-Means is choosing the right value of K. The Elbow Method helps solve this by plotting the **WCSS (Within-Cluster Sum of Squares)** against different values of K. As K increases, WCSS decreases. The optimal K is found at the 'elbow' — where the rate of decrease sharply changes.

> ■ ***Intuition:*** *Imagine you have 100 students. If K=1, all students are in one group (very different from each other — high WCSS). As you increase K, groups become more similar internally. At some point, adding more groups barely helps — that's the elbow!*

K-Means vs K-Medoids

Aspect	K-Means	K-Medoids (PAM)
Center	Mean of cluster points (may not be a real data point)	Actual data point (medoid)
Robustness	Sensitive to outliers	More robust to outliers
Speed	Faster for large datasets	Slower — $O(n^2)$ complexity
Use Case	General clustering, large datasets	When outliers are a concern

Example	Segmenting customers by average spend	Clustering medical patients robustly

The Apriori Algorithm

Apriori is a classic algorithm for mining frequent itemsets and generating association rules. It uses a **bottom-up breadth-first approach**: it starts with individual items and progressively builds larger itemsets, pruning those that fall below a minimum support threshold.

How Apriori Works:

- **Step 1 — Scan:** Count frequency of each individual item (1-itemsets).
- **Step 2 — Prune:** Remove items below minimum support threshold.
- **Step 3 — Join:** Combine frequent items to form 2-itemsets.
- **Step 4 — Repeat:** Continue scanning, pruning, and joining for larger itemsets.
- **Step 5 — Rules:** Generate association rules from frequent itemsets meeting minimum confidence.

■ Apriori Property

Key Principle (Anti-monotone property): If an itemset is infrequent, ALL its supersets must also be infrequent. This lets Apriori skip entire branches of the search space — making it efficient.

The ECLAT Algorithm

ECLAT (Equivalence Class Clustering and bottom-up Lattice Traversal) is a **depth-first** alternative to Apriori. Instead of using horizontal data format, it uses a **vertical data format** where each item stores the list of transaction IDs (TID-list) it appears in.

Aspect	Apriori	ECLAT
Search Strategy	Breadth-first (level by level)	Depth-first (one branch at a time)
Data Format	Horizontal (transactions → items)	Vertical (items → transaction IDs)
Support Count	Scan entire database each pass	Set intersection of TID-lists
Memory	Less memory per pass	Stores TID-lists — can be large
Speed	Slower — multiple DB scans	Faster — no repeated DB scans

Section 3

■ Mathematical Foundations

Euclidean Distance · WCSS · Support · Confidence

1. Euclidean Distance

The Euclidean distance measures the straight-line distance between two points in n-dimensional space. It is the primary distance metric used in K-Means to determine which centroid is nearest to each data point.

■ Formula — Euclidean Distance (2D)

$$d(p, q) = \text{sqrt}((x_2 - x_1)^2 + (y_2 - y_1)^2)$$

In plain English: Subtract the x-coordinates and square the result. Do the same for y-coordinates. Add them and take the square root. This gives you the straight-line distance — just like the Pythagorean theorem!

Example: Point A = (2, 3) and Point B = (5, 7). Distance = sqrt((5-2)² + (7-3)²) = sqrt(9 + 16) = sqrt(25) = **5 units**.

2. Within-Cluster Sum of Squares (WCSS)

WCSS measures how compact the clusters are — how close data points are to their assigned centroid. A lower WCSS indicates tighter, better-defined clusters. K-Means minimizes WCSS during training.

■ Formula — WCSS

$$\text{WCSS} = \text{Sum of [distance(point, centroid)}^2 \text{] for all points in all clusters}$$

In plain English: For every point in every cluster, calculate how far it is from its centroid, square that distance, and add all these squared distances together. The smaller this total, the better the clustering!

3. Support

Support measures how frequently an itemset appears in the dataset. It is expressed as a proportion of all transactions. Only itemsets with support ≥ minimum support threshold are considered 'frequent'.

■ Formula — Support

$$\text{Support(A)} = \text{(Number of transactions containing A) / (Total transactions)}$$

Example: If Bread appears in 60 out of 100 transactions, Support(Bread) = 60/100 = **0.60 or 60%**. If minimum support is 50%, Bread is a frequent item.

4. Confidence

Confidence measures the reliability of an association rule. For rule A → B, confidence tells us: given that A is in a transaction, how often is B also present? Higher confidence means the rule is more reliable.

■ Formula — Confidence

$$\text{Confidence}(A \rightarrow B) = \text{Support}(A \cup B) / \text{Support}(A)$$

Example: Support(Bread ∪ Milk) = 40%, Support(Bread) = 60%. Confidence(Bread → Milk) = 40%/60% = **0.67 or 67%**. This means 67% of the time someone buys bread, they also buy milk.

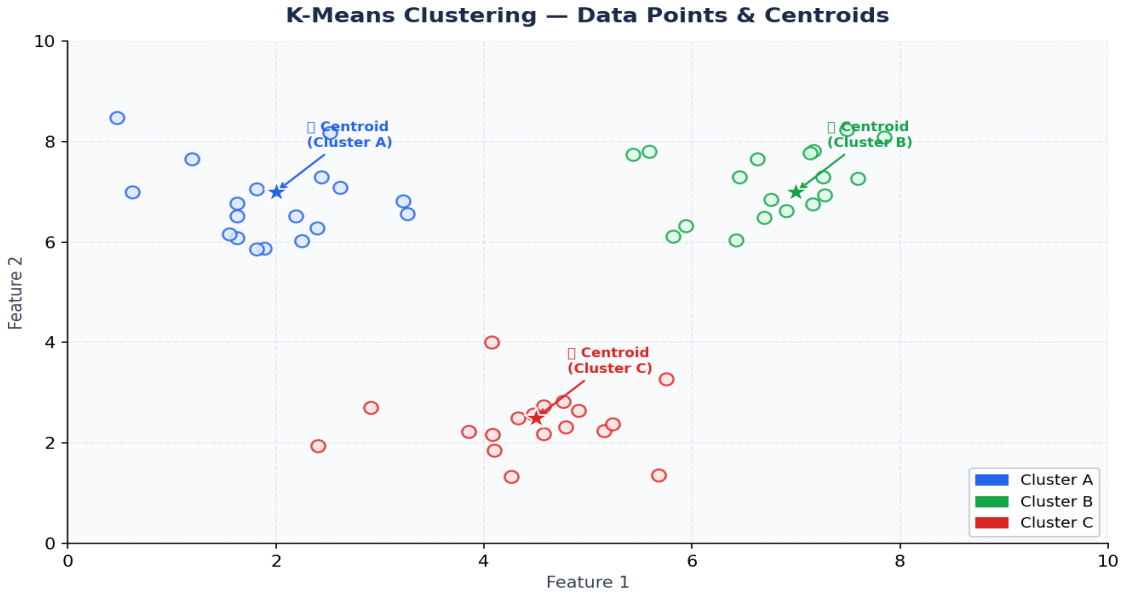
Metric	Meaning	Range	Better When
Euclidean Distance	Straight-line distance between points	0 to ∞	Smaller
WCSS	Total compactness of all clusters	0 to ∞	Smaller
Support	Frequency of itemset in data	0 to 1	Higher
Confidence	Reliability of association rule	0 to 1	Higher

Section 4

■ Visual Diagrams

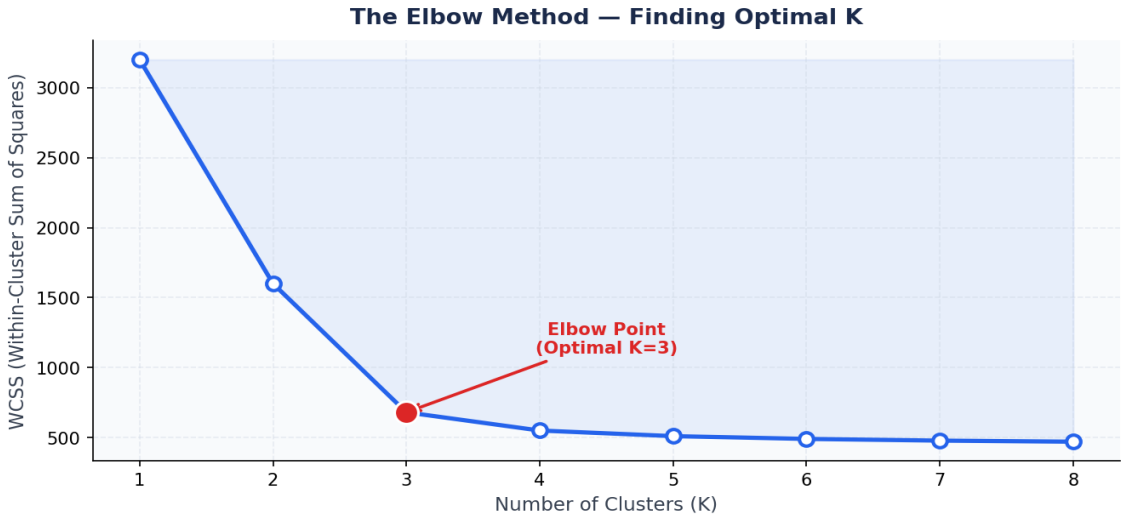
Clustering · Elbow Graph · Market Basket · Data Grouping

Diagram 1: K-Means Clustering with Centroids



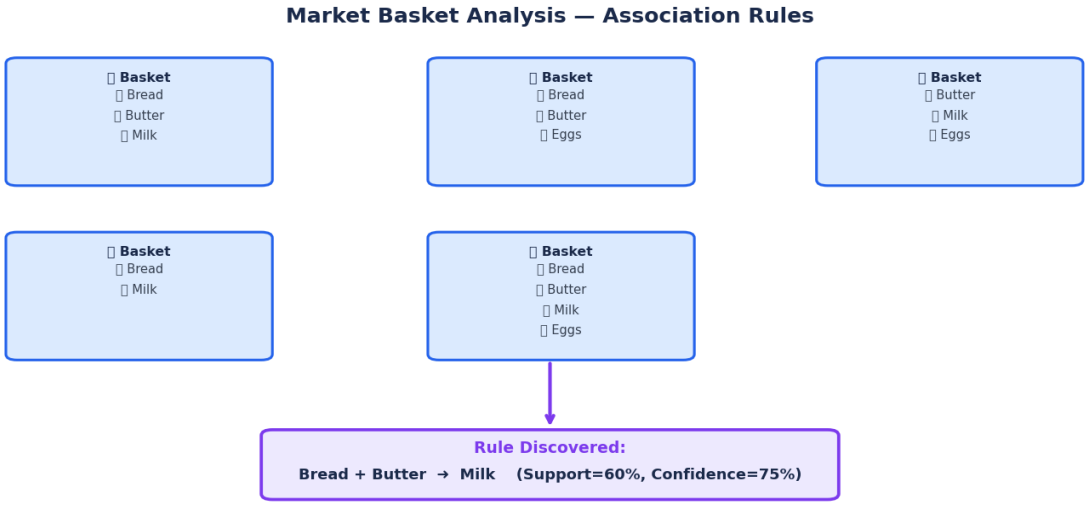
Each cluster is shown in a different color. Stars (■) represent centroids — the computed center of each cluster. Data points near each centroid are assigned to that cluster. K=3 clusters are shown.

Diagram 2: The Elbow Graph (WCSS vs K)



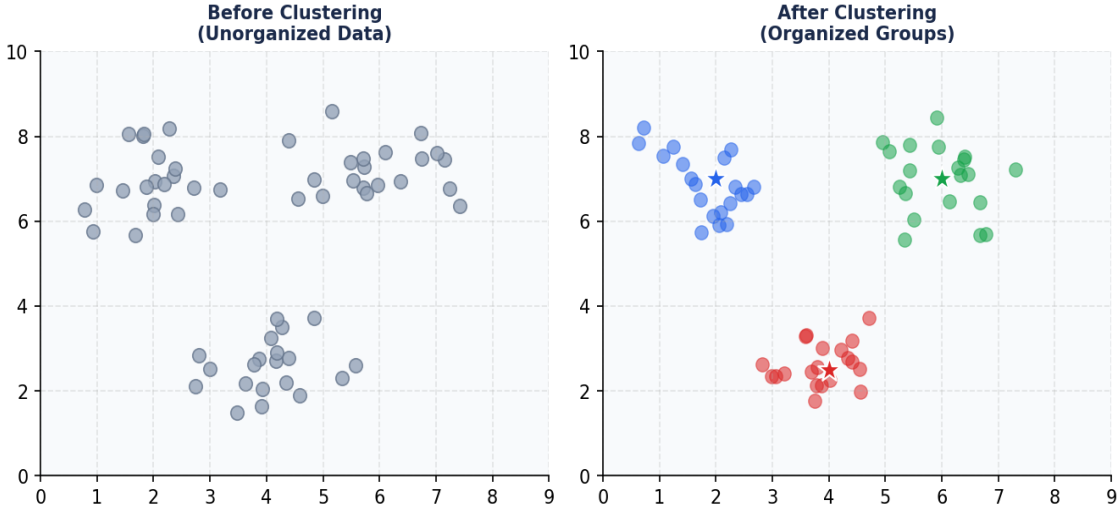
As K increases, WCSS decreases rapidly at first, then levels off. The 'elbow' at K=3 marks the point of diminishing returns — the optimal number of clusters for this dataset.

Diagram 3: Market Basket — Association Rule Discovery



Five shopping baskets are analyzed. Apriori discovers that customers who buy Bread and Butter frequently also buy Milk. The discovered rule has Support=60% and Confidence=75%.

Diagram 4: Data Grouping — Before and After Clustering



Left: Raw, unorganized data with no clear structure. Right: After K-Means clustering, three distinct groups emerge with centroids marked. The algorithm revealed the hidden structure.

Section 5

■■ Workflow

Data Input → Pattern Detection → Clustering / Association Output

The Unsupervised Learning Pipeline

Unsupervised learning follows a structured pipeline from raw data to actionable insights. Each stage transforms the data until meaningful patterns emerge.

■

1. Data Input

Collect raw, unlabeled data. Clean missing values, remove duplicates, normalize features (e.g., scale to 0–1). The quality of input data directly determines the quality of discovered patterns.

■

2. Feature Selection

Choose which attributes (features) to include. Remove irrelevant or redundant features. Dimensionality reduction (e.g., PCA) may be applied to simplify the data.

■
■

3. Algorithm Application

Apply the chosen algorithm: K-Means for clustering, Apriori/ECLAT for association rules. Set hyperparameters (e.g., K value, minimum support threshold).

■

4. Pattern Detection

The algorithm iteratively processes data. K-Means converges when centroids stabilize. Apriori mines frequent itemsets meeting the support threshold. Patterns emerge from repeated iterations.

■

5. Output Generation

Clustering: Each data point is assigned to a cluster. Association: Rules of the form 'IF A THEN B' are generated with support and confidence scores.

■

6. Evaluation & Interpretation

Evaluate cluster quality (Silhouette Score, WCSS). Filter association rules by confidence. Translate findings into business insights and decisions.

Section 6

■ Real-World Applications

Customer Segmentation · Recommendations · Fraud Detection

■ Customer Segmentation

Businesses use clustering to group customers based on purchasing behavior, demographics, or browsing patterns. Each segment receives targeted marketing.

- Retail chains cluster shoppers: budget buyers, luxury buyers, frequent buyers.
- Netflix clusters users by viewing habits to tailor the UI.
- Banks cluster customers for personalized loan and card offers.
- Algorithm: K-Means or K-Medoids on customer feature vectors.

■ Recommendation Systems

Unsupervised learning identifies patterns in user behavior to suggest relevant content — without needing explicit user preferences labeled.

- Spotify discovers music genres automatically from listening data.
- Amazon uses association rules: 'Customers who bought X also bought Y'.
- YouTube clusters videos to recommend similar content.
- Algorithm: Apriori / ECLAT for co-purchase rules; clustering for user grouping.

■ Fraud Detection

Anomaly detection identifies unusual patterns that deviate significantly from normal clusters — flagging potentially fraudulent activity.

- Credit card companies flag transactions far from a user's normal cluster.
- Insurance companies detect fraudulent claims as outliers.
- Cybersecurity uses clustering to detect network intrusion patterns.
- Algorithm: K-Means — outliers are points far from any centroid.

■ Medical Imaging & Diagnostics

Clustering groups patients by symptom patterns or tissue types in images, helping doctors identify disease subtypes without prior labels.

- Grouping cancer patients by tumor gene expression profiles.
- Segmenting brain MRI scans into tissue types (white matter, grey matter).
- Discovering new disease subtypes from patient records.
- Algorithm: K-Means, Hierarchical Clustering on medical data.

■ Topic Modeling & Document Clustering

Clustering groups documents, news articles, or social media posts by topic, enabling automatic organization and search.

- Google News automatically clusters news articles by topic.
- Library systems group academic papers by research area.
- Social media platforms detect trending topics.

- Algorithm: K-Means on TF-IDF feature vectors of documents.

Section 7

■ Quiz

15 Questions — MCQ + Conceptual | Test Your Understanding

■ **Instructions:** Answer all 15 questions. MCQs (Q1–10) have one correct answer each. Conceptual questions (Q11–15) require 1–3 sentence answers. Answers and explanations are in Section 8.

Part A — Multiple Choice Questions (Q1–Q10)

Q1. What is the primary goal of unsupervised learning?

(A) Predict a labeled output for new inputs
(B) Discover hidden patterns or groupings in unlabeled data
(C) Minimize classification error on test data
(D) Memorize training examples for future retrieval

Q2. In K-Means clustering, what does the centroid represent?

(A) The data point furthest from all other points
(B) An actual data point chosen randomly
(C) The mean (average) position of all points in a cluster
(D) The median data point in a sorted cluster

Q3. What does WCSS stand for and what does it measure?

(A) Weighted Cluster Sum Score — accuracy of clusters
(B) Within-Cluster Sum of Squares — compactness of clusters
(C) Within-Class Standard Score — variance of features
(D) Whole Cluster Statistical Score — size of clusters

Q4. Which value of K should you choose using the Elbow Method?

(A) The K where WCSS is highest
(B) Always K = 2, regardless of the graph
(C) The K at the 'elbow' — where WCSS stops decreasing sharply
(D) The largest K value tested

Q5. How does K-Medoids differ from K-Means?

(A) K-Medoids uses the mean of each cluster as the center
(B) K-Medoids is only for binary data
(C) K-Medoids uses an actual data point (medoid) as the cluster center
(D) K-Medoids does not require specifying K in advance

Q6. In the Apriori algorithm, what is the 'anti-monotone property'?

(A) Frequent itemsets can only contain frequent sub-itemsets
(B) If an itemset is infrequent, all its supersets must also be infrequent
(C) The support of an itemset always increases with size
(D) Association rules are always monotone in confidence

Q7. A store has 200 transactions. Bread appears in 120 of them. What is Support(Bread)?

(A) 120
(B) 0.40

(C) 0.60
(D) 200

Q8. Support(Bread ∪ Butter) = 0.40 and Support(Bread) = 0.60. What is Confidence(Bread → Butter)?

(A) 0.24
(B) 0.67
(C) 1.50
(D) 0.40

Q9. Which algorithm uses a vertical data format (TID-lists) instead of a horizontal transaction format?

(A) Apriori
(B) K-Means
(C) ECLAT
(D) K-Medoids

Q10. Which application best demonstrates association rule mining?

(A) Grouping patients by disease severity
(B) Finding frequently purchased product combinations in a supermarket
(C) Detecting spam emails using labeled training data
(D) Predicting house prices from features

Part B — Conceptual Questions (Q11–Q15)

Q11. Explain in your own words why unsupervised learning is called 'unsupervised'. Use an analogy.

Your answer:

Q12. Describe one situation where K-Medoids would be preferred over K-Means. Explain your reasoning.

Your answer:

Q13. A data scientist runs K-Means with K=1 and gets WCSS = 5000. With K=10, WCSS = 200. Does this mean K=10 is always the best choice? Why or why not?

Your answer:

Q14. Explain the difference between Support and Confidence in association rule mining. Why are both metrics needed?

Your answer:

Q15. Name two real-world applications of unsupervised learning and briefly explain how the algorithm is applied in each case.

Your answer:

Section 8

■ Answer Key

Correct Answers with Full Explanations

Q1 — Correct Answer: B

Unsupervised learning works with unlabeled data and aims to discover hidden patterns, groupings, or structures — without any pre-defined correct answers.

Q2 — Correct Answer: C

The centroid is the arithmetic mean of all data points assigned to a cluster. It may or may not coincide with an actual data point.

Q3 — Correct Answer: B

WCSS (Within-Cluster Sum of Squares) measures how compact clusters are by summing the squared distances from each point to its centroid. Lower = tighter clusters.

Q4 — Correct Answer: C

The elbow is where WCSS stops dropping sharply. Beyond this point, adding more clusters gives only marginal improvement but increases complexity.

Q5 — Correct Answer: C

K-Medoids selects an actual data point (medoid) as the cluster representative, making it more robust to outliers than K-Means, which uses the mean.

Q6 — Correct Answer: B

The anti-monotone property means: if {A, B} is infrequent, then {A, B, C} must also be infrequent. Apriori uses this to prune the search space efficiently.

Q7 — Correct Answer: C

Support(Bread) = 120/200 = 0.60 or 60%. Support is always a proportion between 0 and 1.

Q8 — Correct Answer: B

Confidence(Bread→Butter) = Support(Bread∪Butter)/Support(Bread) = 0.40/0.60 ≈ 0.667 or 67%.

Q9 — Correct Answer: C

ECLAT uses a vertical TID-list format. Each item stores the set of transaction IDs it appears in. Support is computed via set intersection — no need to scan the full database.

Q10 — Correct Answer: B

Market basket analysis (finding products bought together) is the canonical application of association rule mining. Options A and C are clustering/supervised tasks respectively.

Conceptual Answers (Q11–Q15)

Q11:

It's 'unsupervised' because there is no supervisor (teacher) providing correct answers. The algorithm learns by itself. Analogy: A librarian sorts thousands of unlabeled books by topic based only on content patterns — no one told them which categories to use.

Q12:

K-Medoids is preferred when the dataset contains outliers. Since K-Medoids uses actual data points (medoids) as centers, extreme outliers don't distort the cluster center. Example: Medical patient clustering where a few extreme cases shouldn't define a cluster.

Q13:

No. While more clusters always reduce WCSS, too many clusters create overfitting — each point becomes its own cluster (WCSS=0 when K=n). The Elbow Method helps find the optimal K that balances compactness with generalizability.

Q14:

Support tells us how often an itemset appears in the data (frequency). Confidence tells us how reliable a rule is (given A, how often does B appear). A high-support rule might have low confidence (common items together by chance). Both metrics are needed to find rules that are both frequent AND reliable.

Q15:

1) Customer Segmentation: K-Means clusters customers by purchase history, demographics, etc. Each segment gets targeted marketing. 2) Recommendation Systems: Apriori mines co-purchase rules (e.g., bread→milk) to recommend products. Both discover hidden patterns without labeled training data.

Section 9

■ Glossary

Key Terms Defined — Quick Reference

Cluster	A group of data points that are more similar to each other than to points in other groups. Clusters reveal natural groupings in data without predefined labels.
Centroid	The geometric center of a cluster — calculated as the arithmetic mean of all data points in the cluster. In K-Means, the centroid is updated each iteration.
Support	A metric in association rule mining that measures how frequently an itemset appears across all transactions. Formula: (transactions with itemset) / (total transactions). Range: 0 to 1.
Confidence	A metric measuring the reliability of an association rule $A \rightarrow B$. It is the conditional probability that B appears given A is present. Formula: $\text{Support}(A \cup B)/\text{Support}(A)$.
WCSS	Within-Cluster Sum of Squares. A measure of cluster compactness — the sum of squared distances between each data point and its cluster centroid. Lower WCSS = tighter, better-formed clusters.
Association Rule	An if-then relationship between items discovered in a dataset, written as $A \rightarrow B$ (if A then B). Used in market basket analysis and recommendation systems.
K-Means	An iterative clustering algorithm that partitions n data points into K clusters by minimizing WCSS. Assigns each point to the nearest centroid, then recalculates centroids.
K-Medoids	A clustering algorithm similar to K-Means, but uses actual data points (medoids) as cluster centers instead of means. More robust to outliers.
Apriori Algorithm	A frequent itemset mining algorithm that uses a breadth-first approach and the anti-monotone property to efficiently discover association rules.
ECLAT Algorithm	Equivalence Class Clustering and bottom-up Lattice Traversal. A depth-first algorithm using vertical TID-lists for faster frequent itemset mining.
Euclidean Distance	The straight-line distance between two points in n-dimensional space. Calculated as the square root of the sum of squared coordinate differences.

Elbow Method	A technique for selecting the optimal K in K-Means by plotting WCSS vs K and identifying the 'elbow' — the point of diminishing returns in WCSS reduction.
Unsupervised Learning	A machine learning paradigm where algorithms discover hidden patterns in unlabeled data without guidance from pre-defined categories or answers.
Medoid	The most centrally located actual data point in a cluster. Used as the cluster representative in K-Medoids. Unlike a centroid, a medoid must be a real data point.
TID-list	Transaction ID List. Used in ECLAT — a list of all transaction IDs in which an item appears. Support is computed by intersecting TID-lists of items.

End of Chapter 2 Study Guide — Unsupervised Learning

Review the diagrams in Section 4, practice the quiz in Section 7, and refer to the glossary as needed.